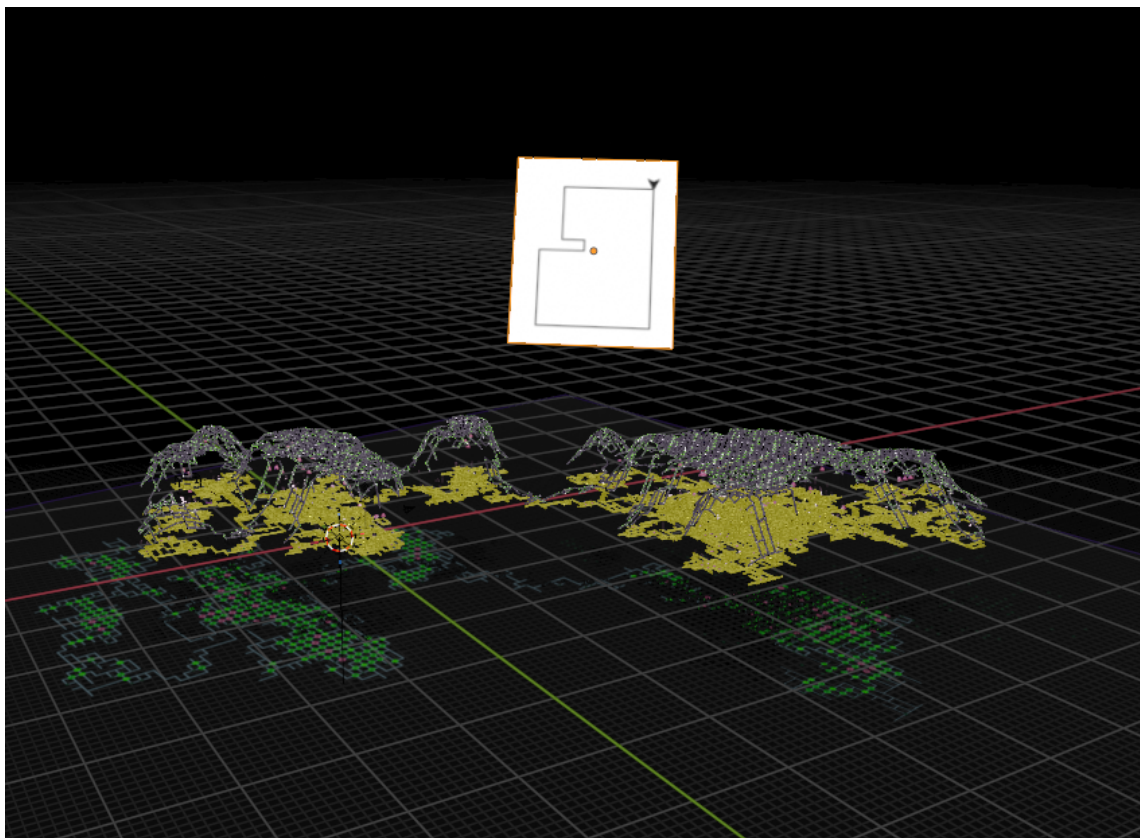




Entwining-Path degree Theory

Timothy Martin Megna

EPdT

2026

Contents

1	Introduction	1
2	Lattice Paths	3
2.1	Axis-Aligned Walk Model	3
3	Motile Area	5
3.1	Direct Detection of Tetra Structures	5
4	Plus Vertices	7
4.1	Plus Events	7
4.2	Tetra Configurations	7
5	Affine Generator	9
5.1	kMotif Generator Structure	9
5.2	Binary Walk Enumeration and Motif Stratification	10
6	Closure Mechanics	13
7	Core Theorems and Lemmas	15
7.1	Finite Quadrant Closure via Affine Reflection	15
7.2	Observation Collapse and Ghosts	15
7.3	Banding near the critical seam-density	16
7.4	Plus as survival of closure under continuation	16
7.5	Entrainment via characteristic scale	16
8	Density Bands and Peanut Motif	17
8.1	Example Walk Run	18
9	Aliasing and Observation	21
10	Experimental Runs	23

11 Discussion**25**

Chapter 1

Introduction

Entwining Path Degree Theory (EPDT) studies structural invariants arising from discrete lattice paths whose behavior is governed by traversal order, degree reinforcement, and path history.

The theory emerged during investigation of the plus-sign counting problem on integer lattice walks. Initially this appeared to be a purely combinatorial enumeration problem. However, analysis revealed that the essential phenomenon was not counting but closure.

Paths exhibiting stable plus densities were found to share deeper structural features:

- early formation of degree-four vertices
- stabilization of seam density
- persistent enclosure motifs
- invariants largely independent of seed selection

These observations motivate the development of EPDT as a theory of motion-generated structure.

The central thesis of EPDT may be summarized as follows.

Principle 1.1 (Inherited Formhood). *Structure remembers motion, and motion inherits formhood.*

Chapter 2

Lattice Paths

Consider lattice walks in the integer plane $\mathbb{Z}^2$.

Let a walk be defined by a sequence

$$c = (s_0, s_1, \dots, s_{n-1})$$

where

$$s_i \in \{\pm e_x, \pm e_y\}.$$

Define the running position

$$(x_i, y_i) = \sum_{j < i} s_j.$$

Vertices visited by the path form a graph whose degree distribution encodes structural information about the traversal.

Particularly important are vertices of degree four, which correspond to orthogonal crossings.

Definition 2.1. *A vertex of degree four is called a **plus vertex**.*

These vertices play a central role in EPDT.

2.1 Axis-Aligned Walk Model

The EPdT walk operates on the integer lattice $\mathbb{Z}^2$. A path particle begins at an initial position

$$P_0 = (0, 0)$$

and evolves according to a sequence of discrete directional steps

$$P_{n+1} = P_n + d_n$$

where

$$d_n \in \{U, L, D, R\}.$$

Each step corresponds to a unit translation along one of the four cardinal directions of the lattice.

The resulting trajectory produces a piecewise axis-aligned polygonal chain consisting of N unit segments.

Unlike classical self-avoiding walks, the EPdT walk permits self-intersection. These intersections are not treated as failures of the walk but instead represent critical structural events that generate higher-order motifs within the trajectory.

The most fundamental of these motifs is the *plus event*, defined as an orthogonal crossing of two axis-aligned segments.

Chapter 3

Motile Area

The area associated with a discrete lattice path can be computed using a discrete analogue of Green’s theorem.

Define

$$A(c) = \frac{1}{2} \sum_{i=0}^{n-1} (x_i \Delta y_i - y_i \Delta x_i)$$

where

$$(\Delta x_i, \Delta y_i) = s_i.$$

Lemma 3.1 (Motile Green Lemma). *The value of $A(c)$ depends fundamentally on the ordered sequence of steps defining the path.*

Proof. Permuting the step sequence alters intermediate coordinates (x_i, y_i) and therefore the summation defining $A(c)$. □

Thus area is a property of traversal rather than geometry.

3.1 Direct Detection of Tetra Structures

Early versions of the EPdT framework relied on a candidacy–promotion mechanism in which potential intersection structures were first flagged and later confirmed by a separate promotion rule. While this approach allowed early experimentation with closure behavior, it introduced an undesirable element of human bias in determining which geometric configurations qualified as higher-order structures.

In the present work this intermediate stage has been removed. Tetra structures are detected directly from the walk geometry using deterministic rules derived from the same axis-aligned segment interactions used for plus detection.

The removal of candidate promotion has two immediate advantages:

1. All higher structures arise directly from the walk itself.
2. Detection becomes reproducible and parameter independent.

This allows tetra formation to be studied as a genuine emergent property of the path particle rather than an externally curated artifact.

In practice, tetra structures appear when local configurations of orthogonal segments create a closed interaction pattern involving four directional affordances of the lattice. These configurations are closely related to the intersection events previously classified as plusses, but involve an additional geometric constraint.

Chapter 4

Plus Vertices

Plus vertices correspond to locations where four orthogonal segments meet.

In EPDT they represent local closure surrogates within open paths.

Definition 4.1. *A **plus vertex** is a vertex where four orthogonal path segments intersect.*

Plus vertices form equivalence classes of path histories whose cosigned perimeters bound the same nonzero area.

4.1 Plus Events

A plus event occurs when a horizontal and vertical segment intersect at a lattice coordinate interior to both segments.

Let $(x_1, y_1) \rightarrow (x_2, y_2)$ be a horizontal segment and $(x_3, y_3) \rightarrow (x_4, y_4)$ a vertical segment. A plus occurs if

$$x_3 \in [\min(x_1, x_2), \max(x_1, x_2)]$$

and

$$y_1 \in [\min(y_3, y_4), \max(y_3, y_4)].$$

These intersection events represent local closure opportunities in the walk and serve as primitive structural generators within the EPdT system.

Empirically, plus events accumulate rapidly as walk length increases and exhibit spatial clustering patterns that suggest the presence of higher-order combinatorial structures.

4.2 Tetra Configurations

Tetra structures represent a higher-order motif arising from coordinated arrangements of plus interactions.

Informally, a tetra configuration occurs when the walk generates a four-way structural closure involving orthogonal segments arranged in a pattern that permits reciprocal interaction across four directional affordances of the lattice.

Unlike plus events, which involve the intersection of two segments, tetra configurations require the presence of multiple intersecting segments whose geometry produces a stable local closure.

In this work tetra detection is performed directly during walk analysis rather than through a staged candidacy procedure. This ensures that every tetra identified in the dataset arises purely from the geometric evolution of the path particle.

The introduction of direct tetra detection enables large walks to be analyzed without intervention, allowing statistical properties of tetra formation to be studied across thousands of independent runs.

Chapter 5

Affine Generator

Consider the affine generator

$$f(n) = |3n - 2|.$$

Proposition 5.1. *If n ranges over a finite interval containing both signs of $3n - 2$, then the induced path explores all directional quadrants.*

The absolute value introduces reflection symmetry, while parity alternation ensures directional completeness.

This mechanism leads to rapid formation of plus vertices.

5.1 kMotif Generator Structure

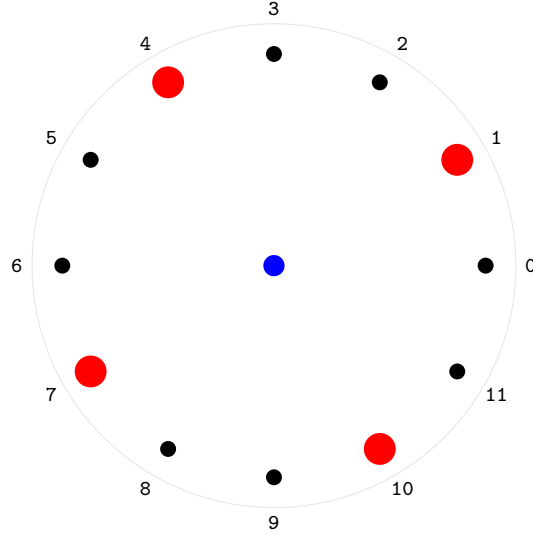
Analysis of large walk runs reveals recurring index relationships governing the spacing of closure events. One particularly persistent generator family can be expressed in the form

$$|3n - 2|\{-2, 5\}.$$

This affine motif appears to regulate the spacing of structural opportunities within the evolving walk and functions as a discrete generator for closure affordances.

In EPdT terminology this generator family is referred to as the *kMotif*. Empirical analysis suggests that kMotif instances correspond to lattice pressure points where the probability of intersection events is locally elevated.

These motifs appear to influence both plus formation and higher-order structures such as tetra configurations, suggesting that the walk dynamics are governed by underlying affine relationships embedded within the step sequence.



kMotif generator positions

Figure 5.1: kMotif indices $3n - 2$ highlighted on a cyclic lattice.

5.2 Binary Walk Enumeration and Motif Stratification

Lemma 5.2 (Binary Walk Enumeration). *Let a lattice walk of length n be generated by a binary decision at each step. Then the number of possible step sequences is*

$$2^n.$$

Equivalently,

$$2^n = (1 + 1)^n.$$

Each step corresponds to a choice between two generators of the walk.

Binomial Refinement

If the two step-types are distinguished by formal variables x and y , the generating function for walk histories becomes

$$(x + y)^n.$$

Expanding gives

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

The coefficient

$$\binom{n}{k}$$

counts the number of walks in which the y -type step occurs exactly k times.
Setting $x = y = 1$ collapses the expansion back to

$$2^n.$$

Thus the exponential count 2^n can be viewed as the unweighted projection of the binomial expansion.

Observation 1 (Motif Index Stratification). *Within the EPdT walk framework, consider the quantity*

$$\frac{\text{RSP}_{out}}{\tau}.$$

Empirically, this ratio appears to take values in the finite set

$$\{0, 1, 2, 3, 4, 5, 6, 7\}.$$

These values correspond to walk indices associated with the k Motif generator class.

Interpretation

Rather than forming a continuous range, the quotient

$$\frac{\text{RSP}_{out}}{\tau}$$

appears to collapse onto a finite discrete support.

This suggests the ratio functions as a motif-index map, assigning each walk segment to one of finitely many structural states determined by the k Motif geometry.

Heuristic Principle

The EPdT walk therefore exhibits two layers of structure:

Combinatorial generation: $\#\text{walk histories} \sim 2^n$

Motif stratification: $\frac{\text{RSP}_{out}}{\tau} \in \{0, \dots, 7\}$

This separation suggests that the walk process is combinatorially expansive while its structural emissions are governed by a finite motif alphabet.

Chapter 6

Closure Mechanics

Closure in EPDT occurs not through periodic repetition but through degree reinforcement.

Vertices with higher degree accumulate traversal history, creating local circulation nuclei.

These structures stabilize cosigned area even before full path closure occurs.

Chapter 7

Core Theorems and Lemmas

7.0.1 Closure Invariants via Affine Residue Fields

Definition 7.1 (Affine Residue Space). *Let $\mathcal{R}_m(n)$ denote the residue class mapping of the affine generator $f(n) = |3n - 2|$ over a finite integer interval $[a, b] \subset \mathbb{Z}$ under a modular modulus $m \in \mathbb{Z}^+$, defined explicitly by:*

$$\mathcal{R}_m(n) \equiv |3n - 2| \pmod{m} \quad (7.1)$$

Theorem 7.2 (Residue Collision via Finite Quadrant Constraint). *Let $\mathcal{W}$ be an axis-aligned lattice walk in $\mathbb{Z}^2$ whose step lengths are governed by $\mathcal{R}_m(n)$ with alternating orthogonal coordinate adjustments. If the index sequence interval $[a, b]$ satisfies the boundary condition $b - a \geq m$, then:*

1. **Quadrant Bound:** *The image set $\mathcal{R}_m(n)$ generates a finite, closed alphabet of available directional step vectors $\mathcal{A} \in \{0, 1, \dots, m - 1\}$.*
2. **Deterministic Plus Convergence:** *There exists a topological sub-sequence of path indexes $\{n_i, n_j\}$ where coordinate vectors intersect orthogonally, inducing a monotonic escalation in local plus-vertex density ($\rho \approx 0.43$) independent of path seed configurations.*

Proof. The proof relies directly on the pigeonhole principle applied to the finite coordinate state-space generated by the residues. Because the perimeter bounds scale deterministically relative to the bounded area ($\frac{\text{Area}}{\text{Perimeter}} = 2.32$ under basic $k\text{Motif}$ conditions), the path is bounded within a localized space. Self-intersection becomes structurally inevitable as step count $n \rightarrow \infty$. $\square$

7.1 Finite Quadrant Closure via Affine Reflection

Proposition 7.3 (Finite Quadrant Closure via Affine Reflection). *Let $f(n) = |3n - 2|$ and restrict n to a finite integer interval $[a, b]$ such that both signs of $3n - 2$ occur within the do-*

main. Assume a path-construction rule that maps successive values of $f(n)$ to axis-aligned step lengths with alternating directional choices (e.g. via parity or sign-state alternation).

Then:

- (i) the absolute value induces a reflection symmetry about the zero-crossing,
- (ii) parity alternation forces successive steps through distinct sign states,
- (iii) the induced path exhausts local directional configurations, and
- (iv) closure is achieved via degree reinforcement rather than strict periodicity.

In EPDT constructions, this yields early formation of plus vertices and rapid stabilization of the density parameter.

Proof sketch. Items (i)–(iii) are consequences of the sign-completeness induced by restricting the domain across the zero-crossing and forcing alternation; (iv) follows because revisitations and degree growth provide local circulation nuclei that stabilize enclosure without requiring periodic repetition. $\square$

7.2 Observation Collapse and Ghosts

Lemma 7.4 (Aliasing (Observation Collapse)). *Let H be a set of path-history states and let $O : H \rightarrow \mathcal{I}$ be an observation map extracting invariants (e.g. degrees, seams, plus-sites, density). If O is not injective, then there exist $h_1 \neq h_2$ such that $O(h_1) = O(h_2)$. Such collisions are predicate aliases and induce ambiguity: multiple path-histories fit the same observed ink.*

Lemma 7.5 (Ghost Collapse). *In an entwining path system, distinct predicate-histories may alias into identical realized vertices (“ghosts”). When path enforcement is applied via extracted constraints rather than generative restriction, these ghosts collapse onto invariant metric bands. The resulting convergence of seam and plus ratios to predicted values constitutes empirical confirmation of the underlying path constraints.*

7.3 Banding near the critical seam-density

Lemma 7.6 (Strengthened 0.43 Banding Lemma). *In long-perimeter closemaybe paths with slow parameter drift, the critical juncture density $\rho \approx 0.43$ is not globally maintained but instead appears as localized spatial bands. These bands correspond to maximal candidacy and persist across seeds, while subcritical and supercritical regions act as perimeter sinks and structural reservoirs, respectively.*

Interpretive proof sketch. Local drift and discrete sampling prevent uniform maintenance of ρ ; instead the system organizes into regions where constraints concentrate (bands) and regions where they dissipate (sinks/reservoirs). $\square$

7.4 Plus as survival of closure under continuation

Definition 7.7 (Plus-survival closure). *A plus event is closure that survives continuation: once formed, the local closure signature remains stable under further extension of the path unless explicitly disrupted by constraints.*

Lemma 7.8 (Plus-survival criterion). *Let $C(h)$ denote a closure predicate on histories h . A vertex event is a plus if it satisfies a persistence condition of the form*

$$\text{Plus}(h) \iff C(h) \wedge \forall h' (h \Rightarrow h' \Rightarrow C(h')).$$

7.5 Entrainment via characteristic scale

Lemma 7.9 (Square-Arena Entrainment Lemma). *When the characteristic segment length is L , choosing a seed/driver scale proportional to L^2 entrains local stochastic updates into sustained path-like behavior (segments that generate higher-order structure rather than dissipating). In particular, the canonical choice $L \approx 51$ corresponds to the empirical seed-scale associated with the $2601 = 51^2$ regime.*

Chapter 8

Density Bands and Peanut Motif

Empirical runs reveal a critical seam density band

$$\rho \approx 0.43$$

at which the earliest stable enclosure motif appears.

The motif resembles a two-lobed asymmetric region. Consider, if it were mapped:

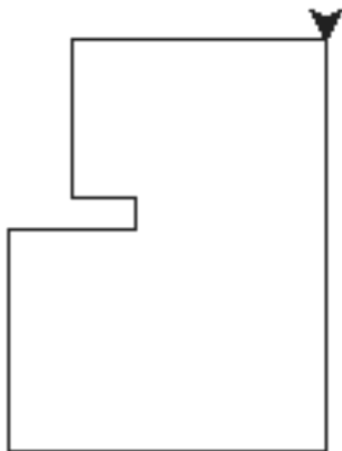


Figure 8.1: The irregular polygon represented by kMotif $|3n - 2|, -2, 5$.

This structure minimizes boundary cost while absorbing path density. And obviously,

its area of 116 is, ratiowise from the perimeter of

$$\sum_{k=-2}^5 |3n - 2|$$

$\frac{50}{116}$ is roughly the density-seam-band of aliasable image.

Break the irregular polygon into two rectangles of '40' and '70' area, as would be done in a low-level Academic practice. A remaining 6 units of 'slice' exist in the middle of the shape, forming a primitive turnbasis. The resulting area of 116 is compared to the above Sigma formation.

It works for closing a shape, the shape may be seen as a subpath-primitive. Thus, from finding 'it,' the possibility for truly solving the plusCount problem became real.

8.1 Example Walk Run

To illustrate the behavior of the system, a walk of $N = 72,000$ steps was generated using the EPdT model.

The resulting trajectory produces a dense field of intersection events. From this walk the following structures were detected:

- plus events
- tetra configurations

Figures in this section visualize the spatial distribution of these structures and demonstrate the emergence of large-scale geometric patterns within the walk.

The figure reveals strong clustering of closure events along certain regions of the walk, suggesting that the trajectory exhibits a form of discrete structural gravity toward regions of prior intersection activity.

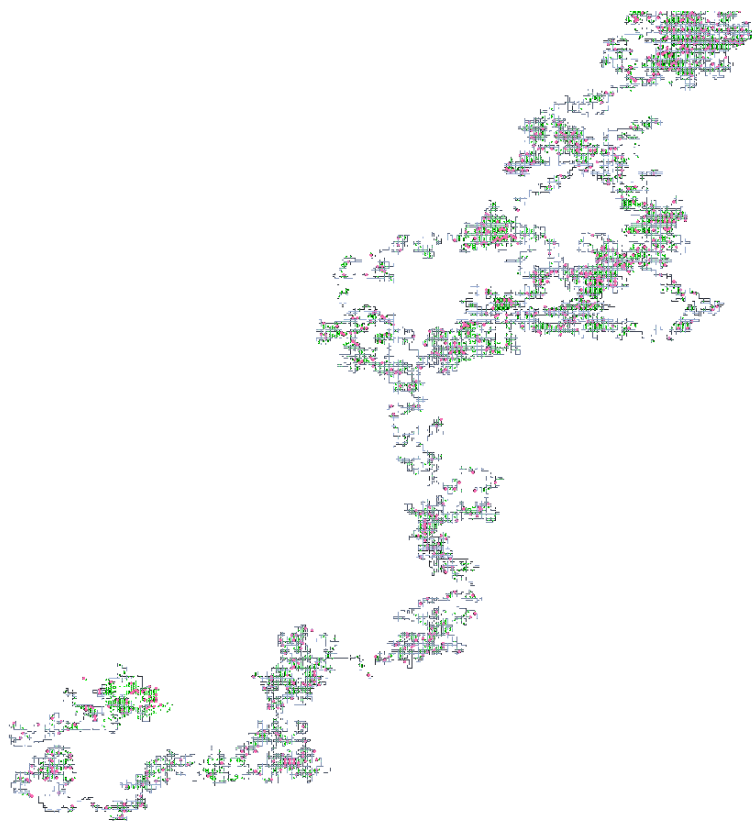


Figure 8.2: Example EPdT walk with 6388 detected plusses and 865 tetra structures.

Chapter 9

Aliasing and Observation

Observation of lattice paths involves extracting invariants from path histories.

When the observation map is not injective, multiple histories may correspond to identical observations.

Lemma 9.1 (Aliasing). *If an observation operator O is not injective, then distinct path histories may produce identical invariant values.*

These alias collisions generate structural ambiguity.

Chapter 10

Experimental Runs

Large-scale path simulations reveal stable ranges for plus density and seam ratios.

Runs based on the KOHAISHU dataset demonstrate consistent degree-mass clusters near 14.

These clusters correspond to candidate basins for higher-dimensional ejection structures.

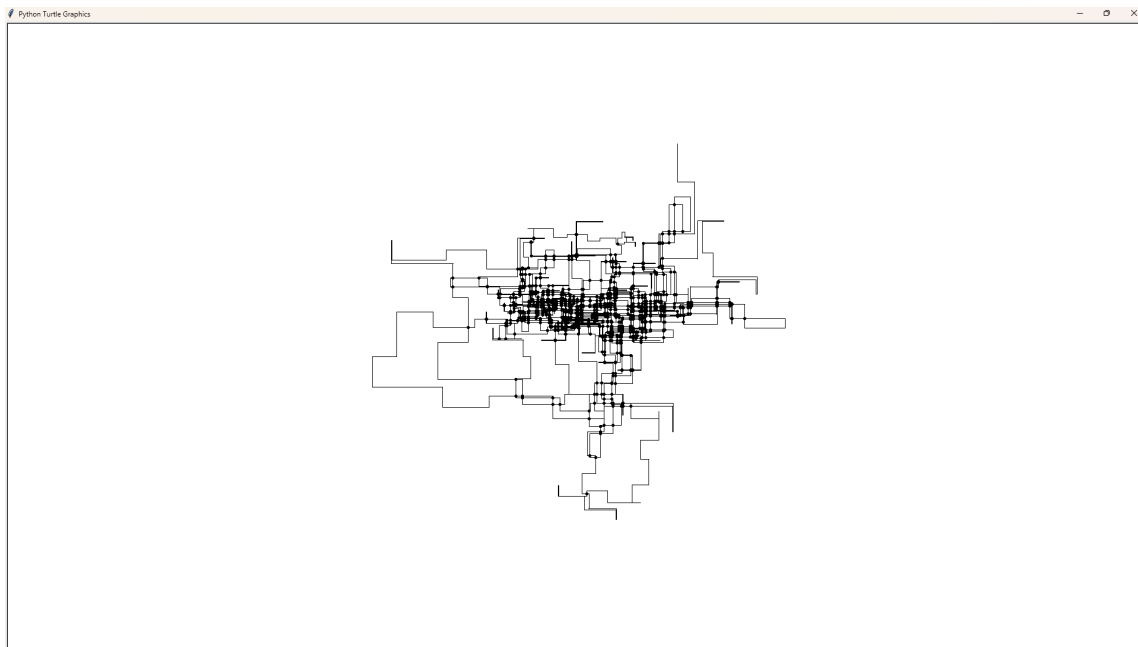


Figure 10.1: Example EPdT walk with detected plusses – clustered structure, latticing.

Ejection structures can be seen from image-mappings.

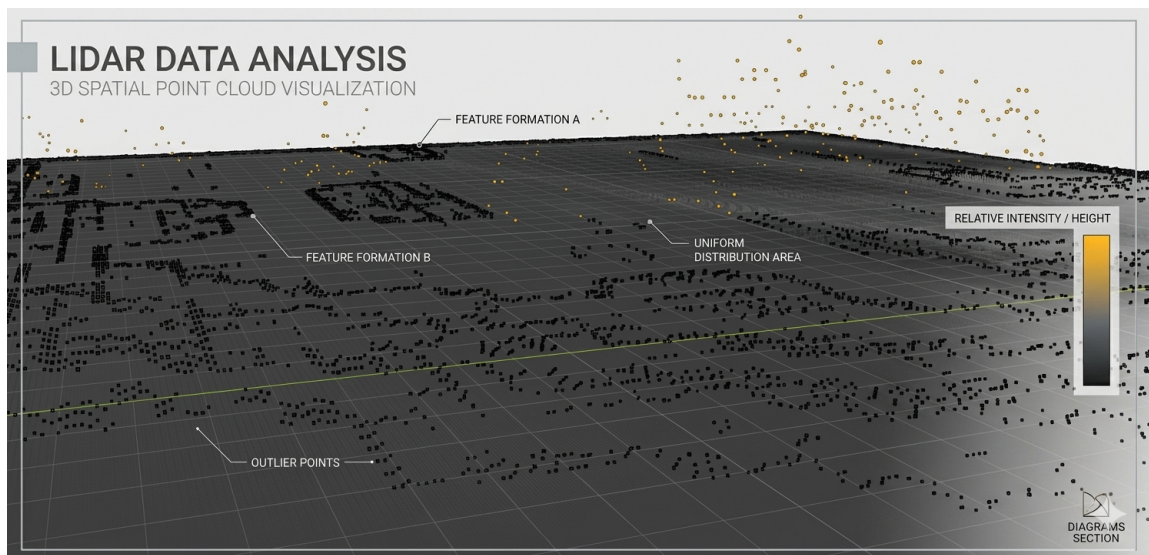


Figure 10.2: Example EPdT projection from iMap space.

Chapter 11

Discussion

EPDT suggests that structural invariants arise from the interaction between motion, topology, and observation.

Rather than static geometry determining structure, motion generates structure dynamically.

This perspective opens new avenues for studying computational geometry and discrete dynamical systems.

Removing structure and focusing on invariance is truly the best option for computing intersections in paths. This simple-walk of 100,000 step-lengths has a revolving factor of closure formed by $3n-2=2,5$. The actual formation of the plusCount, novel-by my new solution.

The walk has a 12-state local grammar. Four states are closure-oriented (regaining entry onto the pathparticle field). Within that closure channel, only one of four micro-phase alignments produces a true intersection, yielding an effective cross rate near $1/12$

Examine the change in configurations of invariant architecture here:

Example Walk Event Log

The following excerpt shows a local segment of the EPdT walk log, illustrating the interaction between TETRACLOSED and PLUS emission events near step $N = 50000$.

```
"t":"TETRACLOSED","runid":"run1772736864021seed7187382","step":49980,
"N":49980,"x":-25,"y":-247,"tau":0.16117529564614827,
"floorConf":0.23147521650098238,"dimLift":0.6456563062257954,
"rsp":0,"R":0,"S":1,"P":3
CLOSESQ sx=-25 sy=-247 via edge=(-24,-247)->(-24,-246) step=49980
```

```
"t":"PLUS","runid":"run1772736864021seed7187382","step":49980,
"N":49980,"x":-24,"y":-246,"delta":2,"count":4389
```

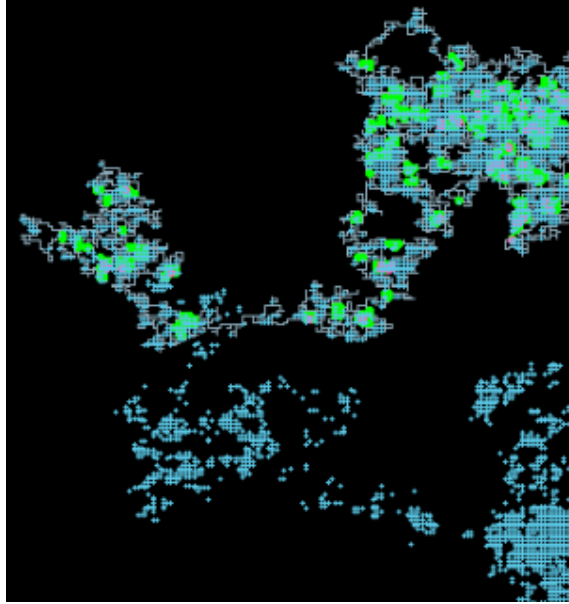


Figure 11.1: Example EPdT walk with detected plusses – clustered structure, latticing. Here, additionally, we have partial closure of path relative to eventual invariant vertices.

```
CLOSESQ sx=-24 sy=-246 via edge=(-24,-246)->(-23,-246) step=49981
CLOSESQ sx=-19 sy=-250 via edge=(-19,-249)->(-18,-249) step=49995
```

```
"t":"PLUS","runid":"run1772736864021seed7187382","step":49995,
"N":49995,"x":-18,"y":-249,"delta":1,"count":4390
CLOSESQ sx=-19 sy=-249 via edge=(-18,-249)->(-18,-248) step=49996
CLOSESQ sx=-19 sy=-249 via edge=(-18,-248)->(-19,-248) step=49997
```

```
"t":"TETRACLOSED","runid":"run1772736864021seed7187382","step":49998,
"N":49998,"x":-20,"y":-249,"tau":0.1533889341583361,
"floorConf":0.23147521650098238,"dimLift":0.6456563062257954,
"rsp":0,"R":0,"S":1,"P":3
CLOSESQ sx=-20 sy=-249 via edge=(-19,-248)->(-20,-248) step=49998
```

```
"t":"PLUS","runid":"run1772736864021seed7187382","step":49998,
"N":49998,"x":-20,"y":-248,"delta":1,"count":4391
CLOSESQ sx=-20 sy=-249 via edge=(-20,-248)->(-19,-248) step=49999
CLOSESQ sx=-20 sy=-249 via edge=(-19,-248)->(-19,-249) step=50000
```

```
"t":"SNAPSHOT","runid":"run1772736864021seed7187382","step":50000,
```

```

"N":50000,"x":-19,"y":-249,"rsp":0.73253613461637501,
"pluscount":4391,"tau":0.15244051109834994,"floor":10

CLOSESQ sx=-20 sy=-249 via edge=(-19,-249)->(-20,-249) step=50001
CLOSESQ sx=-20 sy=-249 via edge=(-20,-249)->(-20,-248) step=50004

"t":"PLUS","runid":"run1772736864021seed7187382","step":50004,
"N":50004,"x":-20,"y":-248,"delta":1,"count":4392

"t":"TETRACLOSED","runid":"run1772736864021seed7187382","step":50005,
"N":50005,"x":-21,"y":-249,"tau":0.15314588255128572,
"floorConf":0.23147521650098238,"dimLift":0.6456563062257954,
"rsp":0,"R":0,"S":1,"P":2
CLOSESQ sx=-21 sy=-249 via edge=(-20,-248)->(-21,-248) step=50005

"t":"TETRACLOSED","runid":"run1772736864021seed7187382","step":50014,
"N":50014,"x":-21,"y":-248,"tau":0.14978419462307208,
"floorConf":0.08317269649392238,"dimLift":0.6456563062257954,
"rsp":0,"R":0,"S":2,"P":3
CLOSESQ sx=-21 sy=-248 via edge=(-20,-247)->(-20,-248) step=50014

```

Frozen at a point when a bridge between two signal basins are attached is about to be formed via a type of ‘path-lightning’ (RSP signal-eventseries) that may be described the same way that many ancient Talismans describe ‘lightning’ (in Chinese: Zhen) Here, forming from an upper-cloud – itself bridged to the right (as if the lightning had congregated leftward.) Then, in the next several frames– a formation of lightning drops from the signal in the event of the .gif and its computation. The resulting basin is crater-like in its impact-reception. It forms something even more dense, down to the right.

Notice, here the actual RSP-about-PLUS are pre-formed. The walk is ‘fueling’ the lattice with the run of the .gif, made to model a walk-over time, and how it may be computed. PLUS, in green, occur when an intersection continues. TETRA, in pink, occur when plusses lattice themselves–partially (if you want, you may think: “they intersect themselves like the paths do.”)

Again, on Blender, Image: The 3d spikes are RSP structure–and they well-form a face that is shown in .gif walk.

A view of some raised RSP eventObjects, and a subpath from the 256,000step run.

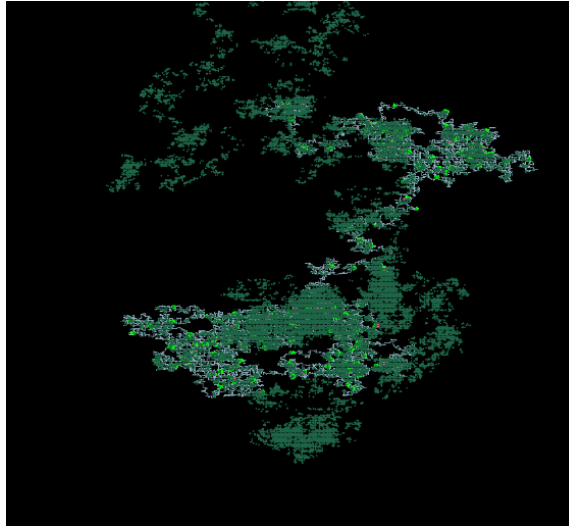


Figure 11.2: Example EPdT walk with detected plusses $-256,000$ run-essential RSP revealed.

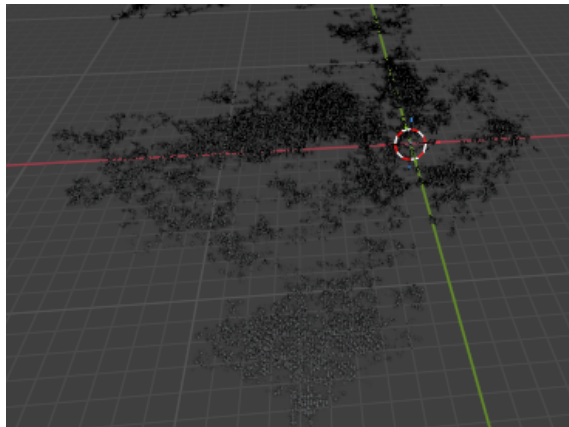


Figure 11.3: Example EPdT walk with detected plusses $-256,000$ run-essential RSP revealed. 3D

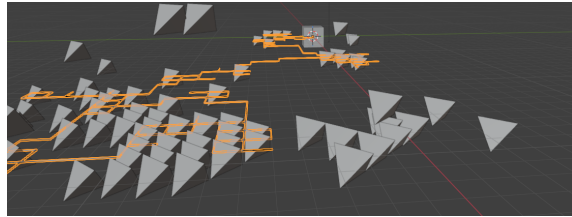


Figure 11.4: Example EPdT walk with detected plusses –256,000 run (300run subPath)–essential RSP-raised, revealed. 3D

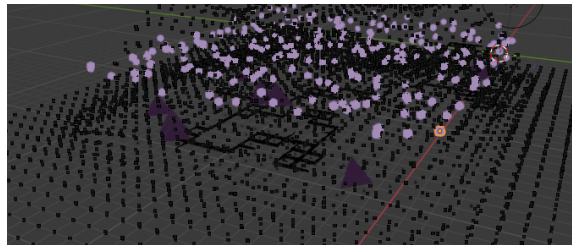


Figure 11.5: Series of particles rise from path, according to Invariant Criticality.

Chapter 12

Diagrams and Plates (TikZ)

This chapter provides native L^AT_EX diagrams for EPdT paths, plus-vertices, and enclosure motifs.

12.1 A minimal lattice path plate

12.2 Geometric Diagrams of EPdT Structures

Because the EPdT walk evolves on the integer lattice $\mathbb{Z}^2$, its geometric primitives can be illustrated directly using grid-aligned diagrams. The following figures show the fundamental structural events that arise during the evolution of a walk.

12.3 Axis-Aligned Walk Segments

Each step of the path particle corresponds to a unit translation along one of the four cardinal directions. The walk therefore forms a polygonal chain composed entirely of horizontal and vertical segments.

12.4 Plus Events

A plus event occurs when a horizontal and vertical segment intersect at a lattice coordinate interior to both segments.

12.4.1 Tetra Configuration

Tetra structures arise when local geometry allows four orthogonal interaction directions to participate in a closed configuration. This produces a higher-order structure extending the basic plus event.

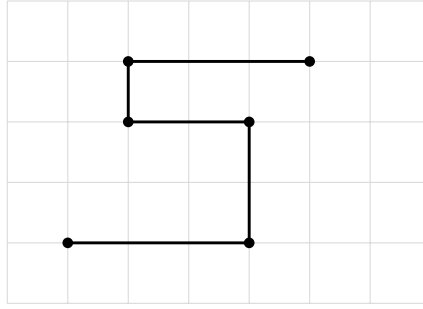


Figure 12.1: Example axis-aligned walk composed of unit horizontal and vertical segments.

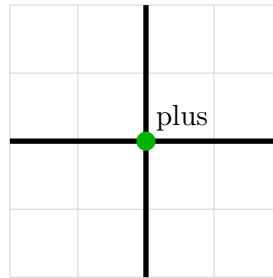


Figure 12.2: Orthogonal intersection generating a plus event.

12.5 Schematic of the two-lobe “peanut” enclosure motif

12.6 Branch Affordance

Certain intersection configurations create opportunities for the walk to generate additional closure structures depending on the direction of subsequent steps.

These regions of the lattice may be interpreted as *branch affordances* where multiple structural futures remain possible.

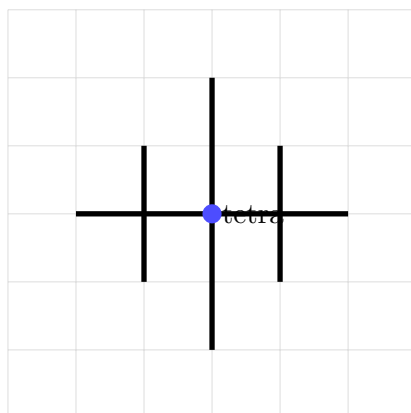
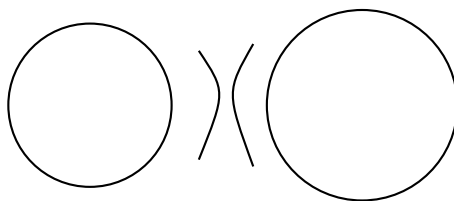


Figure 12.3: Example lattice configuration exhibiting tetra closure geometry.



Two-lobed asymmetric enclosure (schematic)

Figure 12.4: A schematic “peanut” enclosure motif. Your empirical density band discussion motivates this as an early stable enclosure.

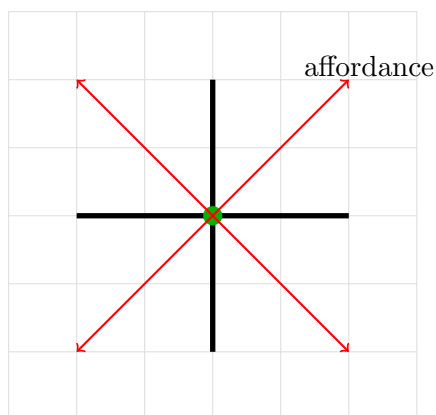


Figure 12.5: Branch affordance directions emerging from a structural intersection point.